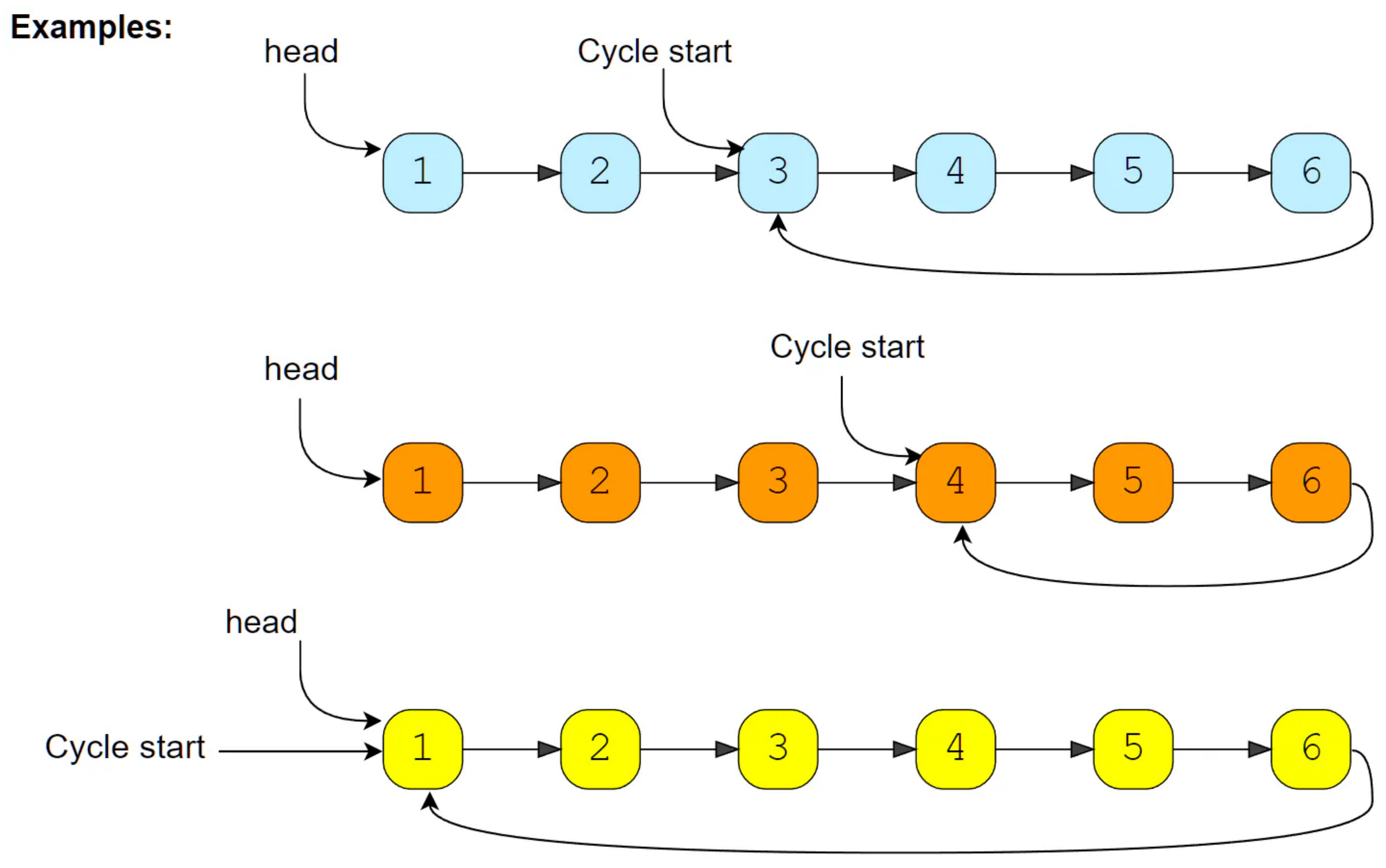
Solution: Start of LinkedList Cycle

Problem Statement

Given the head of a **Singly LinkedList** that contains a cycle, write a function to find the **starting node of the cycle**.

Examples:

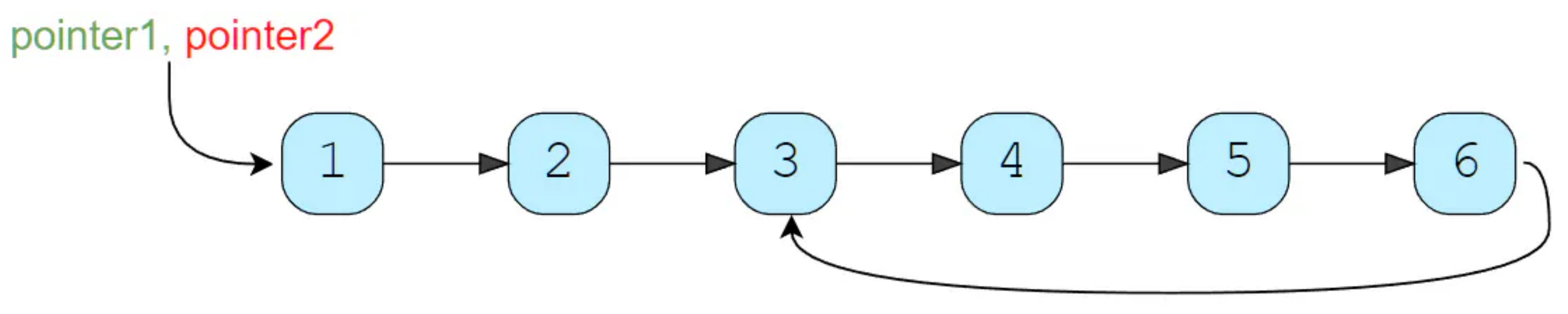


Solution

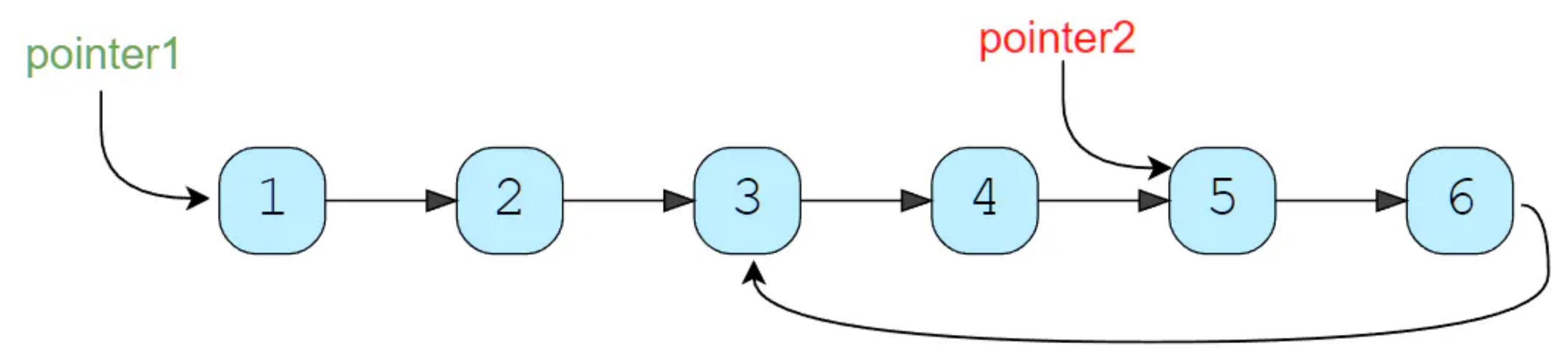
If we know the length of the **LinkedList** cycle, we can find the start of the cycle through the following steps:

1. Take two pointers. Let's call them `pointer1` and `pointer2`.
2. Initialize both pointers to point to the start of the LinkedList.
3. We can find the length of the LinkedList cycle using the approach discussed in LinkedList Cycle. Let's assume that the length of the cycle is 'K' nodes.
4. Move `pointer2` ahead by 'K' nodes.
5. Now, keep incrementing `pointer1` and `pointer2` until they both meet.
6. As `pointer2` is 'K' nodes ahead of `pointer1`, which means, `pointer2` must have completed one loop in the cycle when both pointers meet. Their meeting point will be the start of the cycle.

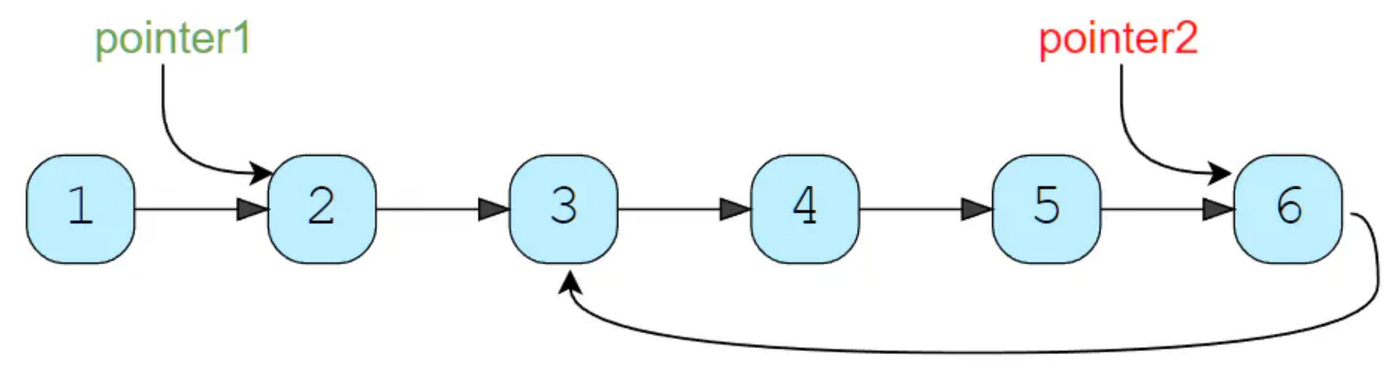
Let's visually see this with the above-mentioned Example-1:

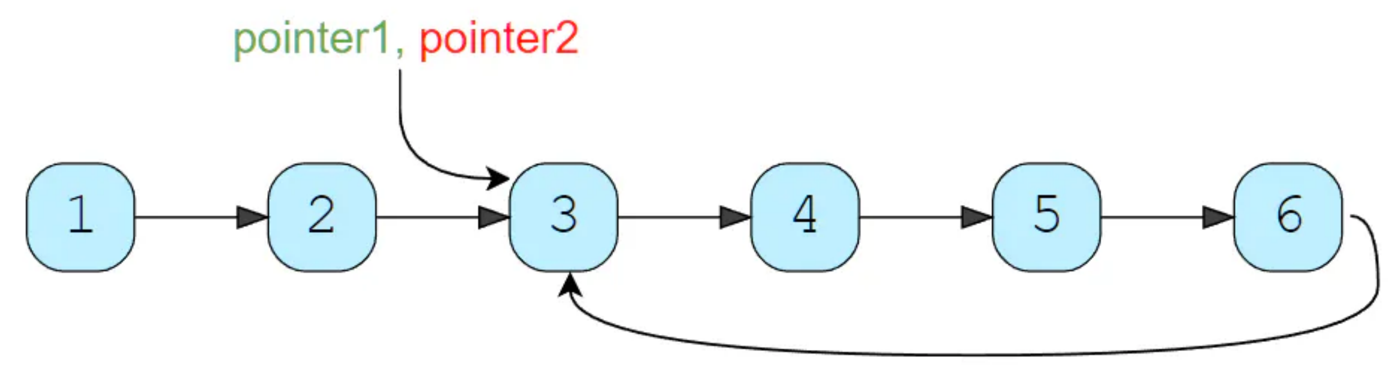


-Move pointer2 '4' nodes ahead



-Keep incrementing both pointers until they meet

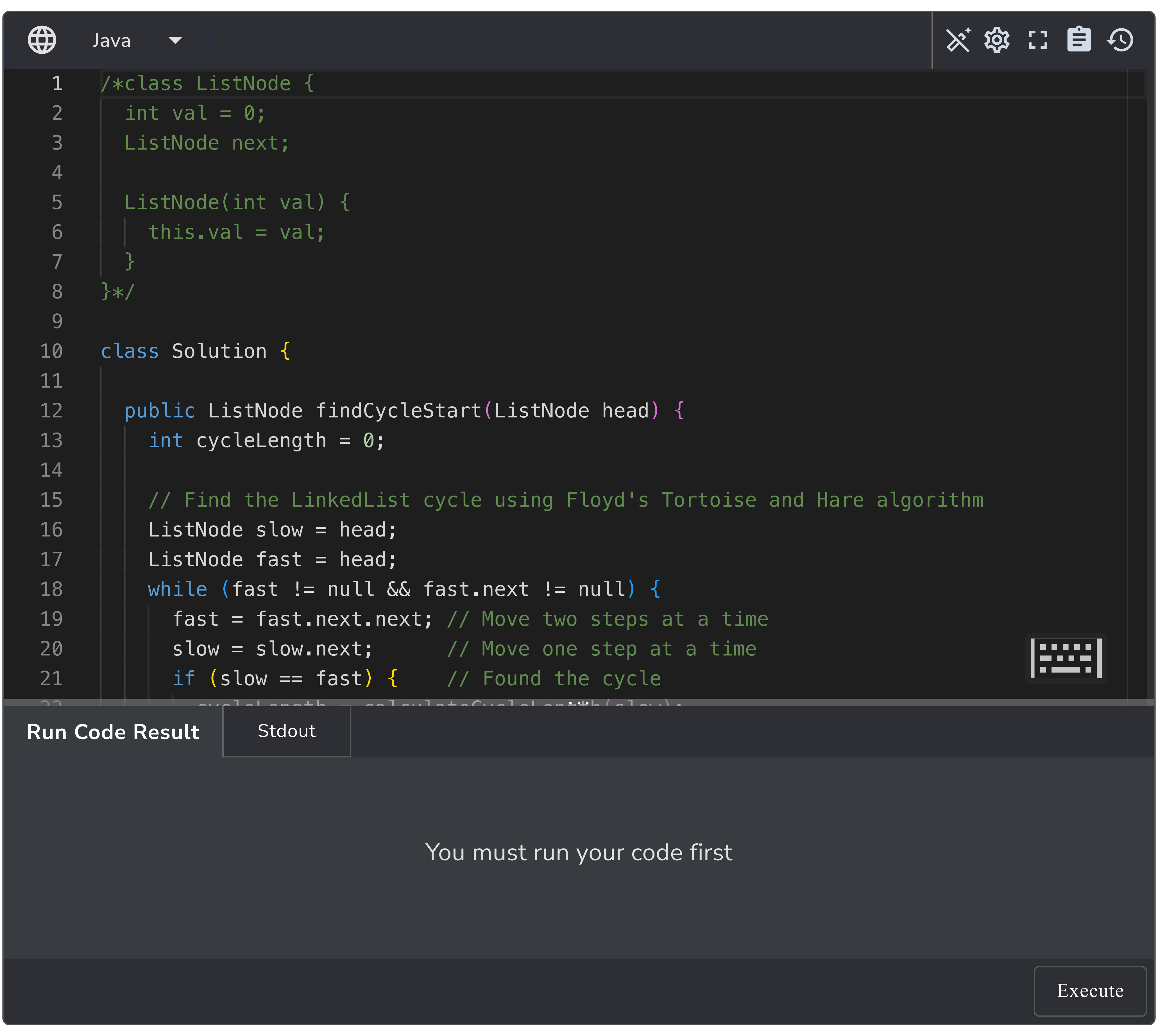




We can use the algorithm discussed in **LinkedList Cycle** to find the length of the cycle and then follow the above-mentioned steps to find the start of the cycle.

Code

Here is what our algorithm will look like:



```java
/*class ListNode {
  int val = 0;
  ListNode next;

  ListNode(int val) {
    this.val = val;
  }
}*/

class Solution {

  public ListNode findCycleStart(ListNode head) {
    int cycleLength = 0;

    // Find the LinkedList cycle using Floyd's Tortoise and Hare algorithm
    ListNode slow = head;
    ListNode fast = head;
    while (fast != null && fast.next != null) {
      fast = fast.next.next; // Move two steps at a time
      slow = slow.next;      // Move one step at a time
      if (slow == fast) {    // Found the cycle
```

You must run your code first

Time Complexity

As we know, finding the cycle in a LinkedList with 'N' nodes and also finding the length of the cycle requires $O(N)$. Also, as we saw in the above algorithm, we will need 4O (N) to find the start of the cycle. Therefore, the overall time complexity of our algorithm will be $O(N)$.

Space Complexity

The algorithm runs in constant space $O(1)$.

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Start Of LinkedList Cycle (Medium)

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Happy Number (Medium)

☐ Mark as Completed